

These are some notes I'm writing on ℓ^2 -Betti numbers (specifically focused on countable discrete groups) with a focus on understanding the cost- ℓ^2 -Betti number problem.

Preliminaries

We start by recalling the construction of the group von Neumann algebra. Let G be a countable discrete group. Then, we may form the space

$$\ell^2(G) = \left\{ (\xi_g)_{g \in G} : \sum_{g \in G} |\lambda_g|^2 < \infty \right\},$$

with orthonormal basis $\{\delta_g\}_{g \in G}$. The left regular representation of G is the map $\lambda: G \rightarrow B(\ell^2(G))$ that takes $\lambda_g \xi(h) = \xi(g^{-1}h)$. The *group von Neumann algebra* is then the WOT-closed linear span of $\lambda(G) \subseteq B(\ell^2(G))$. We will denote this by $L(G)$. Using the right regular representation (as opposed to the left regular representation) $\rho_g \xi(h) = \xi(hg)$ yields the group von Neumann algebra $R(G)$. We have $R(G) = L(G)'$ and $L(G)' = R(G)$. We will usually work with $R(G)$.

The most important fact about the group von Neumann algebra is that it admits a trace, denoted τ_G , defined by

$$\tau_G(T) = \langle x \delta_e, \delta_e \rangle.$$

We will use this trace as part of the construction of the von Neumann dimension. We note that this trace extends to the space $\mathbb{M}_n(R(G)) = R(G) \otimes \mathbb{M}_n(\mathbb{C})$ via the operator $\tau^{(n)} := \tau \otimes \text{Tr}$, where

$$\tau^{(n)}(x) = \sum_{i=1}^n \tau(x_{ii}).$$

We note that the $\ell^2(G)$ can be viewed as a left $\mathbb{C}[G]$ -module by using the *right* G -action,

$$\begin{aligned} G \times \ell^2(G) &\rightarrow \ell^2(G) \\ (g, \xi) &\mapsto \rho_g \xi. \end{aligned}$$

In particular, this means that we may view $R(G)$ as the space of *left* $\mathbb{C}[G]$ -equivariant operators in $B(\ell^2(G))$ — recall that $R(G)$ and $L(G)$ are commutants of each other, so any of the left $\mathbb{C}[G]$ operators will commute with the right action.

The von Neumann Dimension

The first ingredient in understanding ℓ^2 -Betti numbers is the von Neumann dimension.

Definition and Properties

Definition: A Hilbert G -module is a Hilbert space V over $\mathbb{C}$ equipped with a $\mathbb{C}$ -linear isometric left G -action such that there exists $n \in \mathbb{N}$ and an isometric G -equivariant embedding $V \rightarrow \ell^2(G)^{\oplus n}$. Here, it is equivariant with the diagonal left action of G , which we will denote $\lambda^{(n)}$.

Consider a Hilbert G -module H with embedding $i: H \rightarrow (\ell^2(G))^n$. We may identify H with a closed G -invariant subspace of $\ell^2(G)^{\oplus n}$. Let $p_{i(H)} \in B(\ell^2(G)^{\oplus n})^{\lambda^{(n)}}$ be the orthogonal G -equivariant projection onto $i(H)$. We will let $B(H)^G$ be the set of all G -module endomorphisms of H .

Proposition: We claim that the quantity

$$\dim_{R(G)}(H) = \tau^{(n)}(p_{i(H)})$$

is independent of the embedding $i(H)$.

Proof. Suppose $j: H \rightarrow \ell^2(G)^{\oplus m}$ is another embedding with corresponding projection $q_{j(H)}$. Define the map $u: \ell^2(G)^{\oplus n} \rightarrow \ell^2(G)^{\oplus m}$ by taking $u = j \circ i^{-1}$ on $\text{im}(i)$ and 0 on $\text{im}(i)^\perp$. In particular, we have $j = u \circ i$, and $q_{j(H)} = p_{i(H)} \circ u^*$ upon taking adjoints. Therefore, we have

$$\begin{aligned} \tau^{(m)}(q_{j(H)}) &= \tau^{(m)}(j \circ q_{i(H)}) \\ &= \tau^{(m)}(u \circ i \circ q_{i(H)}) \\ &= \tau^{(m)}(u \circ i \circ p_{i(H)} \circ u^*). \end{aligned}$$

Using the tracial property of $\tau^{(m)}$ and exchanging dimension to account for the new range, we have

$$\begin{aligned} &= \tau^{(n)}(i \circ p_{i(H)} \circ u^* \circ u) \\ &= \tau^{(n)}(i \circ p_{i(H)}^2) \\ &= \tau^{(n)}(i \circ p_{i(H)}). \end{aligned}$$

Therefore, the quantity $\dim_{R(G)}(H)$ is well-defined. $\square$

Remark: For maximum generality, we may consider any linear isometric G -embedding into $\ell^2(G) \otimes K$ for any Hilbert space K , but this will not be an issue we will deal with for a while.

The quantity $\dim_{R(G)}(H)$ is called the *von Neumann dimension* of H . Certain authors, such as [LÖ2] and [Kam19] define it slightly differently than the way that [L20] does. However, these are equivalent definitions at the end of the day. We note that this quantity is also always positive, since if $P \in \mathbb{M}_n(R(G))$ is the matrix corresponding to the projection $p_{i(H)}$, then all the diagonal entries P_{jj} of P are positive operators, so since $\tau^{(n)}$ is itself positive, $\dim_{R(G)} = \tau^{(n)}(p_{i(H)})$ is positive.

Hilbert G -modules are quite like standard $\mathbb{C}[G]$ -modules in a variety of ways.

- (i) If $K \subseteq H$ is a closed G -invariant subspace of a Hilbert G -module H , then K is itself a G -module, and we call K a submodule.
- (ii) If $K \subseteq H$ is a closed G -invariant subspace of a Hilbert G -module H , then H/K is also a Hilbert G -module since we may identify it with the closed G -invariant subspace $K^\perp \subseteq H$. We call H/K a quotient module.
- (iii) If H_1 and H_2 are Hilbert G -modules with corresponding embeddings $i_1: H_1 \rightarrow \ell^2(G)^{\oplus n_1}$ and $i_2: H_2 \rightarrow \ell^2(G)^{\oplus n_2}$, then there is an embedding

$$i_1 \oplus i_2: H_1 \oplus H_2 \rightarrow \ell^2(G)^{\oplus(n_1+n_2)},$$

yielding the direct sum Hilbert module.

- (iv) Let H_1 be a Hilbert G_1 -module and H_2 a Hilbert G_2 -module with corresponding embeddings $i_1: H_1 \rightarrow \ell^2(G_1)^{\oplus n}$ and $i_2: H_2 \rightarrow \ell^2(G_2)^{\oplus m}$. These yield a tensor product

$$i_1 \otimes i_2: H_1 \otimes H_2 \rightarrow \ell^2(G_1)^{\oplus n} \otimes \ell^2(G_2)^{\oplus m},$$

where we observe that the codomain is isomorphic to $\ell^2(G_1 \times G_2)^{\oplus mn}$. This yields a tensor product Hilbert $(G_1 \times G_2)$ -module.

- (v) Suppose $G_0 \leq G$ is a finite-index subgroup with index m . There is a functor $\text{Res}_{G_0}^G$ from Hilbert G -modules to Hilbert G_0 -modules obtained by restricting the group action to G_0 .

The corresponding embedding emerges from the fact that taking a system of representatives for the coset space G/G_0 yields a unitary G_0 -equivariant isomorphism $u: \text{Res}_{G_0}^G(\ell^2(G)) \cong \ell^2(G_0)^{\oplus m}$. The restriction $\text{Res}_{G_0}^G(H)$ is a Hilbert G_0 -module.

It turns out, the properties of the von Neumann dimension are similar to the properties of the standard dimension, with only restriction adding an extra layer to the definition.

Theorem: The following properties of the von Neumann dimension hold.

- (i) Normalization: $\dim_{R(G)}(\ell^2(G)) = 1$.
- (ii) Faithfulness: that is, $\dim_{R(G)}(V) = 0$ if and only if $V \cong 0$ as G -modules.
- (iii) Weak Isomorphism Invariance: If $f: V \rightarrow W$ is a Hilbert G -module isomorphism — that is, a left G -equivariant bounded operator satisfying $\ker(f) = 0$ and $\overline{\text{im}}(f) = W$ — then $\dim_{R(G)}(V) = \dim_{R(G)}(W)$.
- (iv) Additivity: consider a weakly exact sequence

$$0 \rightarrow V' \xrightarrow{i} V \xrightarrow{\pi} V'' \rightarrow 0$$

of Hilbert G -modules — that is, i is injective, $\overline{\text{im}}(i) = \ker(\pi)$, and $\overline{\text{im}}(\pi) = V''$. Then, $\dim_{R(G)}(V) = \dim_{R(G)}(V') + \dim_{R(G)}(V'')$.

- (v) Multiplicativity: Let G and K be countable groups, let V be a Hilbert G -module, and let W be a Hilbert K -module. Then, the (Hilbert space) completed tensor product $V \overline{\otimes} W$ is a Hilbert $G \times K$ -module, and satisfies

$$\dim_{R(G \times K)}(V \overline{\otimes} W) = \dim_{R(G)}(V) \dim_{R(K)}(W).$$

- (vi) Restriction: let $G_0 \subseteq G$ be a finite index subgroup, and let V be a Hilbert G -module. Then,

$$\dim_{R(G_0)}(\text{Res}_{G_0}^G(V)) = [G : G_0] \dim_{R(G)}(V)$$

Proof.

- (i) We use the identity map $1 \in B(\ell^2(G))$ for both the embedding and projection, where $\tau(1) = 1$.
- (ii) Notice that both τ on $R(G)$ and Tr on $\mathbb{M}_n(\mathbb{C})$ are faithful, so $\tau^{(n)}(p) = 0$ if and only if $p = 0$.
- (iii) Take the polar decomposition $f = u|f|$, where we have $\ker(u) = \ker(f)$, and u is an isometry. Notice that $|f| = (f^*f)^{1/2} \in B(V)$ is G -equivariant by the assumption that f is G -equivariant and we have

$$\begin{aligned} f^* \lambda_g &= (\lambda_{g^{-1}} f)^* \\ &= (f \lambda_{g^{-1}})^* \\ &= \lambda_g f^*, \end{aligned}$$

so it holds for all polynomials in $f^*f \in B(V)$. In particular, this means that u must also be G -equivariant since the decomposition of f is unique. We will show now that u is a G -equivariant isometry. We have $\ker(u) = \ker(f) = 0$, and since u is a partial isometry, $\text{im}(u) = \overline{\text{im}}(u)$, and since $\overline{\text{im}}(u) = \overline{\text{im}}(f) = W$, it follows that u is G -equivariant.

- (iv) Observe that the von Neumann dimension is additive with respect to direct sums since the trace $\tau^{(n)}$ is additive with respect to direct sums. If $p: V \rightarrow \overline{\text{im}}(i)$ is the orthogonal projection onto the image of i , then there is a weak isomorphism $V \rightarrow \overline{\text{im}}(i) \oplus V''$ of Hilbert G -modules given by $i \mapsto (p(x), \pi(x))$. From (iii), it follows that

$$\dim_{R(G)}(V) = \dim_{R(G)}(\overline{\text{im}}(i) \oplus V'')$$

$$\begin{aligned}
&= \dim_{R(G)}(\overline{\text{im}}(i)) + \dim_{R(G)}(V'') \\
&= \dim_{R(G)}(V') + \dim_{R(G)}(V'').
\end{aligned}$$

- (v) Notice that $\ell^2(G \times K) \cong \ell^2(G) \overline{\otimes} \ell^2(K)$. Since tensor products and direct sums commute, it follows that embeddings $i_1: V \rightarrow \ell^2(G)^{\oplus m}$ and $i_2: W \rightarrow \ell^2(K)^{\oplus n}$ yields an embedding

$$i_1 \otimes i_2: V \otimes W \rightarrow \ell^2(G)^{\oplus m} \overline{\otimes} \ell^2(K)^{\oplus n} \cong \left(\ell^2(G) \overline{\otimes} \ell^2(K) \right)^{\oplus mn} \cong \ell^2(G \times K)^{\oplus mn}.$$

Notice that the trace is multiplicative over tensor products by the definition of the inner product

$$\langle (i_1 \otimes i_2)(v \otimes w), \xi \otimes \eta \rangle = \langle i_1(v), \xi \rangle \langle i_2(w), \eta \rangle,$$

yielding the desired result.

- (vi) Recall from our discussion of the restricted module that we have $\text{Res}_{G_0}^G(\ell^2(G)) \cong \ell^2(G_0)^{\oplus m}$, where $m = [G : G_0]$. We have that the trace τ on $\ell^2(G)$ is then given by $m\tau$ on $\ell^2(G_0)$.

□

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